

NEO Mathematics Learner Guide

How Much Is a Digit Worth? - Place Value

Driving Question: How much is a digit worth?

Your Stepping Stones

Explore how a digit's position changes its value, why placeholder zeros matter, and how neighbouring place-value columns are connected by factors of ten.

A calm route through the lesson

1	Connect: notice the relationship between neighbouring place-value columns.
2	Move: change digits and columns. Predict before checking.
3	Reflect: find a mistake, repair it and explain the misconception.
4	Create: build a Place-Value Museum exhibit that teaches someone else.
5	Rest: when a number feels crowded, focus on one digit and one neighbouring column.

Using the NEO Maths Scratchpad

The Scratchpad is optional. Use Pen, Colour, Width, Eraser, Undo, Redo, Sticky Note and Background. A square-grid or place-value background may help you line up digits. You can save the working page in this browser or save the page as an image.

When you feel stuck: locate the decimal point -> find one digit -> name its column -> state its value -> check one neighbouring column.

Helpful language

Word	Useful idea
digit	one symbol from 0 to 9
place value	the value a digit has because of its position
placeholder zero	a zero that holds a place so other digits keep the correct value
tenfold	ten times as large; neighbouring columns change by a factor of ten
expanded form	a number written as the sum of the values of its digits

Final Task

Curate a Place-Value Exhibition that helps another learner see how position changes digit value. You can use a poster, slide, report or digital exhibition.

Filename pattern: NEO_Maths_PlaceValue_[FirstName]_[Date]

You do not need to complete the lesson in one uninterrupted run. Close panels, pause, use the guide or Scratchpad, and return to one mathematical relationship at a time.